Data Collection & Parameter Definition

While our preliminary proposal considered the CS-VLA standard, applying a 750-850 kg Maximum Take-Off Mass to the HERFUSE horizontal tailplane (HTP) which features a substantial full span of nearly 8 meters creates a severe aerodynamic and structural scale mismatch. Furthermore, the exact aerodynamic loads from the HERFUSE project are omitted due to industrial confidentiality.

To resolve these discrepancies, this project transitions its regulatory framework to the **EASA CS-25 (Certification Specifications for Large Aeroplanes)** standard. We strategically introduce the **ATR 42-600** as our reference regional turboprop aircraft. By utilizing the ATR 42-600's publicly documented mass limits, C.G. envelope, and design speeds, we can accurately synthesize the critical Balancing Tail Loads (BTL) and V-n diagram boundaries without relying on proprietary data.

1. Mass Limits, Aerodynamic Parameters & Sizing Loads

The external loads applied to the APDL model are derived from the critical flight envelope of the ATR 42-600 reference aircraft. Transitioning from the previously proposed Very Light Aeroplane (VLA) standard, these parameters strictly follow EASA CS-25 specifications to ensure aerodynamic validity and precise V-n diagram generation.

Parameter	Value	Role in Structural Analysis & Sizing Loads
Aircraft Category	**Regional Passenger Aircraft** (CS-25 Transport Category)	Shifts the reference framework from a VLA to properly match the substantial physical scale and aerodynamic nature of the HERFUSE HTP baseline.
Sizing Loads Synthesis	**Synthesized V-n Diagram & Balancing Tail Load (BTL)**	Replaces the unavailable proprietary SFBM data. Calculates the ultimate downward tail load using steady-state and dynamic maneuvering equilibrium equations in MATLAB.
Maximum Take-Off Weight (MTOW)	**18,600 kg**	Replaces the 850 kg limit of the CS-VLA standard. Used to determine the maximum lift target (+2.5g) and upper boundaries of the V-n Diagram.

Maximum Zero Fuel Weight (MZFW)	**17,000 kg**	Defines the absolute critical structural condition for wing root bending moments due to the absence of inertial relief (fuel weight).
Center of Gravity (C.G.) Limits	**Forward Limit at 16.8% MAC**	This extreme forward C.G. generates the maximum nose-down pitching moment. It is the critical load case defining the ultimate downward balancing load the HTP must endure.
Limit Manoeuvring Load Factors	$\mathbf{n_1 = +2.5}$ and $\mathbf{n_2 = -1.0}$	Replaces the CS-VLA limits of +3.8 and -1.5. Governed by EASA CS-25.337 to construct the realistic commercial flight envelope bounds.
Main Wing & Tail Dimensions	S_W = **54.5 m²** MAC ($\bar{c}$) = **2.285 m** Tail Moment Arm (l_t) = **11.46 m**	Fundamental geometric variables used in the linearized aerodynamic matrix to calculate moment equilibrium around the C.G..
Design Airspeeds (CS-25)	$v_{s1} \approx$ **110-120 KIAS** V_C/V_{MO} = **250 KIAS**	v_{s1} initiates the aerodynamic lift limit. The Dive Speed (VD) bounds the right edge of the envelope, creating the maximum dynamic pressure (q) and thus the maximum negative BTL for sizing tail spars.
Factor of Safety (FS)	**Minimum 1.5**	A constant constraint applied to all ultimate loads during the APDL structural

		optimization loop to guarantee safety.

2. Horizontal Tail Plane (HTP) Geometry (HERFUSE & ATR 42 Integration)

To guarantee that the APDL scripting and meshing processes remain numerically stable, the exact geometric dimensions and internal structural layout will strictly replicate the original Palumbo et al. (2026) baseline. However, critical aerodynamic properties missing from the literature, which are necessary for the MATLAB equilibrium equations, are integrated using the ATR 42-600 HTP profile.

Parameter	Value	Source & Engineering Application
Airfoil Section	**NACA 0012 (Symmetrical)**	**ATR 42-600:** Ensures linear lift without an intrinsic pitching moment. Essential for MATLAB load synthesis.
Zero-lift Pitching Moment (C_{m0})	**0**	**ATR 42-600:** A constant used in the mathematical equilibrium equations to calculate the exact balancing tail load.
Span	**3978 mm (Semi-span)**	**Literature:** Determines the overall length of the FEA model.
Root Chord	**3090 mm**	**Literature:** Governs the root attachment area dimensions.
Tip Chord	**1860 mm**	**Literature:** Controls the taper ratio of the HTP.
Dihedral Angle	**7 degrees**	**Literature:** Defines the out-of-plane

		geometric setup in APDL.
Max Thickness-to-Chord Ratio	**12%**	**Literature:** Dictates the internal space available for ribs and spars.
Internal Structural Layout	**14 Ribs, 3 Spars, 8 Stringers**	**Literature:** Parametrically defined via APDL ***DO** loops. This serves as the Baseline Configuration (Configuration A) for the structural optimization process.

3. Material Properties

The material selection strategy follows the Palumbo et al. (2026) methodology by combining a robust metallic internal framework with equivalent isotropic composite skins. To ensure maximum structural strength and stiffness under high stress levels, **Aluminum 7075-T6** is selected for the internal framework based on aerospace design standards.

Material / Component	Property	Value	Source & Engineering Application
Aluminum 7075-T6 (Internal framework: Ribs, Spars, Stringers)	Young's Modulus (EX)	**71.7 GPa**	**ASM International :** Governs the elastic deformation and structural stiffness in the APDL linear static analysis.
	Poisson's Ratio (PRXY)	**0.33**	**ASM International :** Required for standard APDL material input (MP,PRXY).
	Tensile Yield Strength	**503 MPa**	**ASM International :** Provides high yield strength, serving as the primary structural threshold for the Factor of Safety (FS > 1.5) constraint.
	Fatigue	**159 MPa**	**ASM International :** Standard

	Strength		high-strength aviation alloy rating for load-bearing members.
Equivalent Isotropic (Composite Skins)	Young's Modulus (EX)	**50.0 GPa**	**Lundsgaard-Larsen et al. (2012):** Based on a quasi-isotropic carbon/epoxy laminate representation. Simulates global skin stiffness efficiently.
	Poisson's Ratio (PRXY)	**0.28**	**Lundsgaard-Larsen et al. (2012):** Standard transverse contraction ratio for quasi-isotropic laminates.
	Density (DENS)	**1600** kg/m^3	**Lundsgaard-Larsen et al. (2012):** Used if inertia or mass calculations are included in the APDL evaluation.